

SOC612_L2

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SOC612: Data Analytics for the Social Sciences

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Lecture 2: Working with data

For the use of external data, we'll check and set a working directory, and import the data from the World Value Survey (Waves 6-7). We assign it the object name 'data'.

For .csv files, we can use `read_csv()` from `readr` (or `read.csv()` from Base-R). For common alternative formats you can use, for example: `read_delim()` for .tsv files (from `readr`) , `read_excel` for .xlsx files (from `readxl`), `readRDS()` for native R files, `fromJSON()` for a JSON string (from `jsonlite`), or `read_xml()` for xml/html data (from `xml2`).

For files from other statistical programmes (e.g., SPSS, SAS, or Stata), you can use `read_sav()` for SPSS, `read_dta()` for Stata, and `read_sas()` for SAS (all in `haven`).

```
library(readr)
```

```
## Warning: package 'readr' was built under R version 4.2.3
```

```
getwd() #manually check for the working directory
```

```
## [1] "C:/Users/raphi/Dropbox/PC/Desktop/LingnanU/SOCI612 Lecture 2"
```

```
setwd(getwd()) #use getwd() inside setwd() to automatically assign the path
data <- read_csv("wvs6_7.csv")
```

```
## Rows: 97220 Columns: 613
## — Column specification —————
## Delimiter: ","
## chr  (12): version, doi, B_COUNTRY_ALPHA, C_COW_ALPHA, LNGE_ISO, X002_02B, V...
## dbl  (601): A_WAVE, A_YEAR, A_STUDY, B_COUNTRY, C_COW_NUM, D_INTERVIEW, S007,...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Wave 6-7 from WVS has 613 variables. For our analysis, we are only interested in a few selected variables. Let's subset the full data into a simpler object. For our analysis on people's view on economic competition (Q106-Q111) and science and technology (Q158-Q163) in East Asian societies, and how they correlate with socio-demographics (Q260, Q262, Q273, Q275). We'll retain the original object. In R, the `tidyverse` package offers many necessary editing functions.

After subsetting the selected variables, we do some checks on variable levels. While subsetting, we'll rename our selected variables.

```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.2.3
```

```
## Warning: package 'ggplot2' was built under R version 4.2.3
```

```
## Warning: package 'tibble' was built under R version 4.2.3
```

```
## Warning: package 'tidyr' was built under R version 4.2.3
```

```
## Warning: package 'purrr' was built under R version 4.2.3
```

```
## Warning: package 'dplyr' was built under R version 4.2.3
```

```
## Warning: package 'forcats' was built under R version 4.2.3
```

```
## Warning: package 'lubridate' was built under R version 4.2.3
```

```
## — Attaching core tidyverse packages — tidyverse 2.0.0 —
## ✓ dplyr      1.1.3      ✓ purrr      1.0.2
## ✓ forcats    1.0.0      ✓ stringr    1.5.0
## ✓ ggplot2    3.5.1      ✓ tibble     3.2.1
## ✓ lubridate  1.9.3      ✓ tidyr      1.3.0
## — Conflicts — tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(magrittr) #for pipe function %>%
```

```
##
## Attaching package: 'magrittr'
##
## The following object is masked from 'package:purrr':
##
##   set_names
##
## The following object is masked from 'package:tidyr':
##
##   extract
```

```
subset <- select(data, country=B_COUNTRY, sex=Q260, age=Q262,  
                 marital_status=Q273, education=Q275, #socio-demographics  
                 eco1=Q106, eco2=Q107, eco3=Q108, eco4=Q109, eco5=Q110, eco6=Q111,  
                 #economic values  
                 sci1=Q158, sci2=Q159, sci3=Q160, sci4=Q161, sci5=Q162, sci6=Q163  
                 #science & technology values  
)  
  
#limit to East Asian societies: China (156), Hong Kong SAR (446), Macao SAR (344), Japan  
(392), South Korea (410), Taiwan ROC (158)  
subset_EA <- subset %>%  
  filter(country %in% c(156, 446, 344, 392, 410, 158))
```

We have limited our dataset to 17 columns, and 9,955 rows, down from 613 columns and 97,220 rows. Let's continue with making it more handy to work with.

- Assign the correct variable levels: Some variables (country, sex, marital status, education) should be ordinal rather than metric/quasi-metric.
- Check the value ranges of the variables. Recode them to NA, where necessary.

```

#countries
subset_EA <- subset_EA %>%
  mutate(country = recode(
    country,
    `156` = "China",
    `446` = "Hong Kong SAR",
    `344` = "Macao SAR",
    `392` = "Japan",
    `410` = "South Korea",
    `158` = "Taiwan ROC"
  )) %>%
  mutate(country = as.factor(country))

#The same can also be done in Base-R:
#subset_EA$country <- factor(
#  subset_EA$country,
#  levels = c(156, 446, 344, 392, 410, 158),
#  labels = c("China", "Hong Kong SAR", "Macao SAR", "Japan", "South Korea", "Taiwan RO
#C"))

#sex
subset_EA <- subset_EA %>%
  mutate(
    sex = recode(
      sex,
      `1` = "male",
      `2` = "female",
      .default = NA_character_
    )
  ) %>%
  mutate(sex = as.factor(sex))

#marital status
subset_EA <- subset_EA %>%
  mutate(
    marital_status = recode(
      marital_status,
      `1` = "married",
      `2` = "married_cohabiting",
      `3` = "divorced",
      `4` = "separated",
      `5` = "widowed",
      `6` = "single",
      .default = NA_character_
    )
  ) %>%
  mutate(marital_status = as.factor(marital_status))

#assume we only care if someone is currently partnered (1,2) or single (3,4,5,6).
subset_EA <- subset_EA %>%
  mutate(
    marital_status = case_when(
      marital_status %in% c("married", "married_cohabiting") ~ "partnered",
      marital_status %in% c("divorced", "separated", "widowed", "single") ~ "single",

```

```

    TRUE ~ NA_character_
  )
) %>%
mutate(marital_status = as.factor(marital_status))

#education
subset_EA <- subset_EA %>%
  mutate(
    education = case_when(
      education %in% c(0, 1) ~ "primary_or_below",
      education %in% c(2, 3) ~ "secondary",
      education %in% c(4, 5) ~ "post_secondary",
      education %in% c(6, 7, 8) ~ "tertiary",
      TRUE ~ NA_character_
    )
  ) %>%
  mutate(education = as.factor(education))

#age
print(range(subset_EA$age, na.rm = TRUE)) #range goes below 18

```

```
## [1] -5 95
```

```

subset_EA <- subset_EA %>%
  mutate(age = ifelse(age < 18, NA, age)) #replace those <18 with NA

#economic values variables: Q106-Q110 (columns 6-10) should fall within a range of 1-10
subset_EA <- subset_EA %>%
  mutate(across(6:10, ~ifelse(between(., 1, 10), ., NA)))

#same for science & technology variables: Q158-Q163 (columns 12-17)
subset_EA <- subset_EA %>%
  mutate(across(12:17, ~ifelse(between(., 1, 10), ., NA)))

#or both in one:
#subset_EA <- subset_EA %>%
#  mutate(across(c(6:10,12:17), ~ ifelse(between(., 1, 10), ., NA)))

#eco6 is a binary factor:
subset_EA <- subset_EA %>%
  mutate(
    eco6 = recode(
      eco6,
      `1` = "environment",
      `2` = "growth",
      .default = NA_character_
    )
  ) %>%
  mutate(eco6 = as.factor(eco6))

```

Assume we want to align the first four of the economic values variables and all six of the science and technology variables along a spectrum. For economic values, 'more collectivist' (1) vs 'more individualist' (10), for science and technology, 'more cautious' (1) vs 'more innovative' (10).

For now, we also just want to work with complete cases on these variables, so we'll remove cases with NA's on them first.

- Remove NA's on eco1-eco4, and sci1-sci6
- eco2 and eco4 have to be reversed.
- sci3, sci4, and sci5 have to be reversed.

In a second step, we want to standardize them. We will do two versions of it:

- We standardize them using the whole sample.
- We standardize them country-wise.

```

#remove NA's
subset_EA <- subset_EA %>%
  filter(complete.cases(eco1, eco2, eco3, eco4,
                        sci1, sci2, sci3, sci4, sci5, sci6))

#alternatively use column numbers
#subset_EA <- subset_EA %>%
# filter(complete.cases(select(., 6:9, 12:17)))

#economic values reversed
subset_EA <- subset_EA %>%
  mutate(across(c(eco2, eco4), ~ 11 - .x))

#science and technology values reversed
subset_EA <- subset_EA %>%
  mutate(across(c(sci3, sci4, sci5), ~ 11 - .x))

#scaling variables (m=0, sd=1)
subset_EA <- subset_EA %>%
  mutate(eco_sum = rowSums(select(., eco1, eco2, eco3, eco4), na.rm = TRUE)) %>%
  mutate(eco_sum = scale(eco_sum)[, 1])

#scaling variables (m=0, sd=1)
subset_EA <- subset_EA %>%
  mutate(sci_sum = rowSums(select(., sci1, sci2, sci3, sci4, sci5, sci6), na.rm = TRUE))
%>%
  mutate(sci_sum = scale(sci_sum)[, 1])

#we also construct country-wise scaled variables
#for the economic values
subset_EA <- subset_EA %>%
  group_by(country) %>%
  mutate(
    eco_sum_cw = rowSums(across(c(eco1, eco2, eco3, eco4), as.numeric), na.rm = TRUE),
    eco_sum_cw = scale(eco_sum_cw)[, 1]
  ) %>%
  ungroup()

#for the science and technology values
subset_EA <- subset_EA %>%
  group_by(country) %>%
  mutate(
    sci_sum_cw = rowSums(across(c(sci1, sci2, sci3, sci4, sci5, sci6), as.numeric), na.rm
= TRUE),
    sci_sum_cw = scale(sci_sum_cw)[, 1]
  ) %>%
  ungroup()

```

In some cases, we might want to create composite indices which combine multiple variables. This can, however, complicate interpretations sometimes, and should usually only be done when it's theoretically well justified, or if it follows an exploratory analysis. Let's assume we did that for our example, and we want to create an Socio-economic index variable based on age, income, and marital status.

- Using age and education level, we create an additive index.

- We weigh it by marital status.

```
subset_EA <- subset_EA %>%
  mutate(
    edu_num = as.numeric(factor(education, levels = c("primary_or_below", "secondary", "post_secondary", "tertiary"))),

    marital_weight = ifelse(marital_status == "partnered", 1.0, 0.8),

    index = (age + edu_num) * marital_weight
  )
```

In some cases, we might want to run multi-level models that involve structural variables besides individual level. Let's assume, for all of the countries in our dataset, we want to add the life expectancy (in 2017), based on gender. The World Bank has a data repository on many structural indicators.

- We load the second dataset (lifeexp.xlsx) into our environment. Since the working directory is already set, we can load the data directly.
- We add the life-expectancies based on gender and country using a `left_join()`.

```
## Warning: package 'readxl' was built under R version 4.2.3
```

There are cases where we might have to create our own specific function, because there is no built-in function from Base-R and no packages contain it. Let's construct the following functions:

- Square function: a function that exponentiates a value by power 2 (2); we'll use this to create an age square variable.
- Percentage function: a function that calculates the percentage of an observed value from the maximum value; we'll use this to create a variable that indicates how much a person has reached in terms of country- and gender-specific life expectancy.
- Conditional function: a function that indicates whether someone's country-specific standardized `eco_sum_cw` (economic values) and `sci_sum_cw` (science and technology) values exceeds the overall values, based on the following criteria: if the country-specific value is higher than the overall value, we assign a 1; if the country-specific value is more than twice as high as the overall value, we assign it a 2; for all other cases, we give them a 0.

```

# 1. square function
sqf <- function(x) {
  return(x^2)
}

subset_EA <- subset_EA %>%
  mutate(age_sq = sqf(age))

# 2. percentage function
prct <- function(part, total) {
  share <- (part / total) * 100
  return(share)
}

subset_EA <- subset_EA %>%
  mutate(age_pct = prct(age, life_exp))

# 3. conditional values function
scaled_values_comp <- function(country_scaled, overall_scaled) {
  ifelse(
    abs(country_scaled) < abs(overall_scaled), 0,
    ifelse(
      abs(country_scaled) > 2 * abs(overall_scaled), 2, 1
    )
  )
}

subset_EA <- subset_EA %>%
  mutate(eco_scaled_comp = scaled_values_comp(eco_sum_cw, eco_sum),
         sci_scaled_comp = scaled_values_comp(sci_sum_cw, sci_sum)
  )

#multiple conditions can, for example, also be written with a case_when() function
#subset_EA$eco_scaled_comp <- case_when(
#  abs(subset_EA$eco_sum_cw) > 2 * abs(subset_EA$eco_sum) ~ 2,
#  abs(subset_EA$eco_sum_cw) < abs(subset_EA$eco_sum) ~ 0,
#  TRUE ~ 1
#)

```

If we want to iterate over a vector or list or another sequence of items, we can also use loops in R. Let's assume we want to mark the positive and negative extreme cases among the respondents. We identify those rows, where the respondents have either the full scores in both economic values (eco1-eco4) and science and technology values (sci1-sci6).

- We create a variable indicating whether a respondent scores 100 ("positive"), 10 ("negative") or any other number ("mixed").

```
vars <- c("eco1", "eco2", "eco3", "eco4",  
         "sci1", "sci2", "sci3", "sci4", "sci5", "sci6")  
  
subset_EA$outlier <- rep(NA_character_, nrow(subset_EA))  
  
for (i in 1:nrow(subset_EA)) {  
  sum <- sum(subset_EA[i, vars], na.rm = TRUE)  
  
  if (sum == 100) {  
    subset_EA$outlier[i] <- "pos"  
  } else if (sum == 10) {  
    subset_EA$outlier[i] <- "neg"  
  } else {  
    subset_EA$outlier[i] <- "mixed"  
  }  
}  
  
subset_EA$outlier <- as.factor(subset_EA$outlier)
```

Before getting into more complex exploratory analyses, let's do a quick descriptive check of the variables and the dataset to see whether all of our changes appear plausible. We have several options for simple descriptive functions, among them `summary()` and `str()` from Base-R, `describe()` from Hmisc, or `glimpse()` from tidyverse/tibble.

```
#Base-R  
summary(subset_EA)
```

```

##      country          sex          age          marital_status
## Length:9216      Length:9216      Min.    :18.00      partnered:6309
## Class :character  Class :character  1st Qu.:33.00      single   :2881
## Mode  :character  Mode  :character  Median :46.00      NA's     : 26
##                                     Mean   :46.25
##                                     3rd Qu.:59.00
##                                     Max.   :95.00
##                                     NA's    :218
##      education      eco1      eco2      eco3
## post_secondary : 895      Min.    : 1.000      Min.    : 1.000      Min.    : 1.000
## primary_or_below:1167      1st Qu.: 5.000      1st Qu.: 4.000      1st Qu.: 3.000
## secondary      :4137      Median : 6.000      Median : 6.000      Median : 5.000
## tertiary       :2962      Mean   : 6.097      Mean   : 5.576      Mean   : 5.266
## NA's           : 55      3rd Qu.: 8.000      3rd Qu.: 7.000      3rd Qu.: 7.000
##                                     Max.   :10.000      Max.   :10.000      Max.   :10.000
##
##      eco4      eco5      eco6      sci1
## Min.    : 1.000      Min.    : 1.000      environment:5510      Min.    : 1.000
## 1st Qu.: 6.000      1st Qu.: 2.000      growth       :3102      1st Qu.: 7.000
## Median : 7.000      Median : 4.000      NA's         : 604      Median : 8.000
## Mean   : 7.124      Mean   : 4.485                                     Mean   : 7.783
## 3rd Qu.: 9.000      3rd Qu.: 6.000                                     3rd Qu.: 9.000
## Max.   :10.000      Max.   :10.000                                     Max.   :10.000
##                                     NA's     :13
##      sci2      sci3      sci4      sci5
## Min.    : 1.000      Min.    : 1.000      Min.    : 1.000      Min.    : 1.000
## 1st Qu.: 6.000      1st Qu.: 4.000      1st Qu.: 4.000      1st Qu.: 6.000
## Median : 8.000      Median : 6.000      Median : 6.000      Median : 8.000
## Mean   : 7.551      Mean   : 5.522      Mean   : 5.896      Mean   : 7.254
## 3rd Qu.: 9.000      3rd Qu.: 7.000      3rd Qu.: 8.000      3rd Qu.: 9.000
## Max.   :10.000      Max.   :10.000      Max.   :10.000      Max.   :10.000
##
##      sci6      eco_sum      sci_sum      eco_sum_cw
## Min.    : 1.000      Min.    :-4.12810      Min.    :-4.51351      Min.    :-4.5278
## 1st Qu.: 6.000      1st Qu.: -0.63020      1st Qu.: -0.83612      1st Qu.: -0.5850
## Median : 8.000      Median : -0.01293      Median : -0.07528      Median : -0.1230
## Mean   : 7.588      Mean   : 0.00000      Mean   : 0.00000      Mean   : 0.0000
## 3rd Qu.: 9.000      3rd Qu.: 0.60435      3rd Qu.: 0.68556      3rd Qu.: 0.6059
## Max.   :10.000      Max.    : 3.27921      Max.    : 2.33405      Max.    : 3.5970
##
##      sci_sum_cw      edu_num      marital_weight      index
## Min.    :-4.86805      Min.    :1.000      Min.    :0.8000      Min.    :15.20
## 1st Qu.: -0.71476      1st Qu.:2.000      1st Qu.:0.8000      1st Qu.:34.00
## Median : -0.06969      Median :2.000      Median :1.0000      Median :47.00
## Mean   : 0.00000      Mean   :2.617      Mean   :0.9373      Mean   :46.35
## 3rd Qu.: 0.63854      3rd Qu.:4.000      3rd Qu.:1.0000      3rd Qu.:59.00
## Max.    : 4.49014      Max.    :4.000      Max.    :1.0000      Max.    :97.00
##                                     NA's     :55      NA's     :26      NA's     :275
##      life_exp      age_sq      age_pct      eco_scaled_comp
## Min.    :75.03      Min.    : 324      Min.    : 20.55      Min.    :0.0000
## 1st Qu.:79.70      1st Qu.:1089      1st Qu.: 40.45      1st Qu.:0.0000
## Median :80.80      Median :2116      Median : 57.09      Median :1.0000
## Mean   :81.70      Mean   :2395      Mean   : 56.78      Mean   :0.6285
## 3rd Qu.:85.70      3rd Qu.:3481      3rd Qu.: 71.97      3rd Qu.:1.0000

```

```
## Max. :87.61 Max. :9025 Max. :117.57 Max. :2.0000
## NA's :1 NA's :218 NA's :218
## sci_Scaled_comp outlier
## Min. :0.0000 mixed:9215
## 1st Qu.:0.0000 pos : 1
## Median :0.0000
## Mean :0.5892
## 3rd Qu.:1.0000
## Max. :2.0000
##
```

```
str(subset_EA)
```

```
## tibble [9,216 × 30] (S3: tbl_df/tbl/data.frame)
## $ country      : chr [1:9216] "China" "China" "China" "China" ...
## $ sex          : chr [1:9216] "female" "female" "male" "male" ...
## $ age          : num [1:9216] 38 69 48 70 30 27 60 63 69 68 ...
## $ marital_status : Factor w/ 2 levels "partnered","single": 1 2 1 1 1 2 1 1 1 1 ...
## $ education     : Factor w/ 4 levels "post_secondary",...: NA 3 NA 4 4 4 3 3 3 3 ...
## $ eco1          : num [1:9216] 4 1 7 5 8 10 1 3 2 2 ...
## $ eco2          : num [1:9216] 6 1 7 4 8 10 9 6 2 2 ...
## $ eco3          : num [1:9216] 1 1 8 1 8 10 1 6 3 2 ...
## $ eco4          : num [1:9216] 10 10 8 10 3 10 8 9 8 10 ...
## $ eco5          : num [1:9216] 1 1 2 1 7 1 9 2 2 1 ...
## $ eco6          : Factor w/ 2 levels "environment",...: 1 1 1 1 1 1 2 2 1 1 ...
## $ sci1          : num [1:9216] 9 10 10 10 8 10 5 9 6 9 ...
## $ sci2          : num [1:9216] 9 10 10 10 4 10 5 5 10 9 ...
## $ sci3          : num [1:9216] 8 1 8 1 6 4 6 5 6 2 ...
## $ sci4          : num [1:9216] 8 1 9 1 3 4 1 5 6 9 ...
## $ sci5          : num [1:9216] 8 1 10 10 9 10 9 8 2 2 ...
## $ sci6          : num [1:9216] 8 10 10 10 8 10 8 9 10 10 ...
## $ eco_sum       : num [1:9216] -0.63 -2.276 1.222 -0.836 0.604 ...
## $ sci_sum       : num [1:9216] 1.066 -1.0897 1.9536 0.0515 -0.4557 ...
## $ eco_sum_cw    : num [1:9216] -0.5 -2.008 1.197 -0.689 0.631 ...
## $ sci_sum_cw    : num [1:9216] 0.574 -1.86 1.577 -0.572 -1.144 ...
## $ edu_num       : num [1:9216] NA 2 NA 4 4 4 2 2 2 2 ...
## $ marital_weight : num [1:9216] 1 0.8 1 1 1 0.8 1 1 1 1 ...
## $ index         : num [1:9216] NA 56.8 NA 74 34 24.8 62 65 71 70 ...
## $ life_exp      : num [1:9216] 80.6 80.6 75 75 75 ...
## $ age_sq        : num [1:9216] 1444 4761 2304 4900 900 ...
## $ age_pct       : num [1:9216] 47.2 85.6 64 93.3 40 ...
## $ eco_scaled_comp: num [1:9216] 0 0 0 0 1 0 0 2 0 0 ...
## $ sci_Scaled_comp: num [1:9216] 0 1 0 2 2 0 1 2 2 2 ...
## $ outlier       : Factor w/ 2 levels "mixed","pos": 1 1 1 1 1 1 1 1 1 1 ...
```

```
#Hmisc
```

```
library(Hmisc)
```

```
## Warning: package 'Hmisc' was built under R version 4.2.3
```

```
##  
## Attaching package: 'Hmisc'
```

```
## The following objects are masked from 'package:dplyr':  
##  
##   src, summarize
```

```
## The following objects are masked from 'package:base':  
##  
##   format.pval, units
```

```
Hmisc::describe(subset_EA)
```

```
## subset_EA
##
## 30 Variables      9216 Observations
## -----
## country
##      n missing distinct
##    9216      0      6
##
## Value      China Hong Kong SAR      Japan      Macao SAR
## Frequency      2919      987      842      2002
## Proportion      0.317      0.107      0.091      0.217
##
## Value      South Korea      Taiwan ROC
## Frequency      1245      1221
## Proportion      0.135      0.132
## -----
## sex
##      n missing distinct
##    9215      1      2
##
## Value      female      male
## Frequency      4914      4301
## Proportion 0.533 0.467
## -----
## age
##      n missing distinct      Info      Mean      Gmd      .05      .10
##    8998      218      76      1      46.25      18.38      21      24
##      .25      .50      .75      .90      .95
##      33      46      59      67      71
##
## lowest : 18 19 20 21 22, highest: 89 91 92 93 95
## -----
## marital_status
##      n missing distinct
##    9190      26      2
##
## Value      partnered      single
## Frequency      6309      2881
## Proportion      0.687      0.313
## -----
## education
##      n missing distinct
##    9161      55      4
##
## Value      post_secondary primary_or_below      secondary      tertiary
## Frequency      895      1167      4137      2962
## Proportion      0.098      0.127      0.452      0.323
## -----
## eco1
##      n missing distinct      Info      Mean      Gmd      .05      .10
##    9216      0      10      0.98      6.097      2.6      2      3
##      .25      .50      .75      .90      .95
##      5      6      8      9      10
##
```

```

## Value      1      2      3      4      5      6      7      8      9     10
## Frequency  457   336   653   585  1420  1290  1662  1671   496   646
## Proportion 0.050 0.036 0.071 0.063 0.154 0.140 0.180 0.181 0.054 0.070
##
## For the frequency table, variable is rounded to the nearest 0
## -----
## eco2
##      n missing distinct      Info      Mean      Gmd      .05      .10
##   9216      0      10    0.978    5.576    2.504      2      3
##    .25    .50    .75    .90    .95
##      4      6      7      8      9
##
## Value      1      2      3      4      5      6      7      8      9     10
## Frequency  448   358   947  1070  1356  2198   918  1031   442   448
## Proportion 0.049 0.039 0.103 0.116 0.147 0.238 0.100 0.112 0.048 0.049
##
## For the frequency table, variable is rounded to the nearest 0
## -----
## eco3
##      n missing distinct      Info      Mean      Gmd      .05      .10
##   9216      0      10    0.986    5.266    2.815      1      2
##    .25    .50    .75    .90    .95
##      3      5      7      8      9
##
## Value      1      2      3      4      5      6      7      8      9     10
## Frequency  817   616  1021   976  1544  1126  1135  1162   392   427
## Proportion 0.089 0.067 0.111 0.106 0.168 0.122 0.123 0.126 0.043 0.046
##
## For the frequency table, variable is rounded to the nearest 0
## -----
## eco4
##      n missing distinct      Info      Mean      Gmd      .05      .10
##   9216      0      10    0.976    7.124    2.305      3      4
##    .25    .50    .75    .90    .95
##      6      7      9     10     10
##
## Value      1      2      3      4      5      6      7      8      9     10
## Frequency  137    77   331   480   735  1661  1407  1887  1147  1354
## Proportion 0.015 0.008 0.036 0.052 0.080 0.180 0.153 0.205 0.124 0.147
##
## For the frequency table, variable is rounded to the nearest 0
## -----
## eco5
##      n missing distinct      Info      Mean      Gmd      .05      .10
##   9203     13      10    0.985    4.485    2.872      1      1
##    .25    .50    .75    .90    .95
##      2      4      6      8      9
##
## Value      1      2      3      4      5      6      7      8      9     10
## Frequency  1432  1013  1270   977  1350   929   950   719   255   308
## Proportion 0.156 0.110 0.138 0.106 0.147 0.101 0.103 0.078 0.028 0.033
##
## For the frequency table, variable is rounded to the nearest 0
## -----
## eco6

```

```

##          n missing distinct
##      8612      604         2
##
## Value      environment      growth
## Frequency      5510      3102
## Proportion      0.64      0.36
## -----
## sci1
##          n missing distinct      Info      Mean      Gmd      .05      .10
##      9216         0         10      0.964      7.783      2.044         5         5
##      .25      .50      .75      .90      .95
##          7         8         9         10         10
##
## Value          1         2         3         4         5         6         7         8         9         10
## Frequency      65      44      143      200      731      807      1533      2272      1265      2156
## Proportion 0.007 0.005 0.016 0.022 0.079 0.088 0.166 0.247 0.137 0.234
##
## For the frequency table, variable is rounded to the nearest 0
## -----
## sci2
##          n missing distinct      Info      Mean      Gmd      .05      .10
##      9216         0         10      0.97      7.551      2.196         4         5
##      .25      .50      .75      .90      .95
##          6         8         9         10         10
##
## Value          1         2         3         4         5         6         7         8         9         10
## Frequency      66      59      201      319      895      1017      1428      2106      1144      1981
## Proportion 0.007 0.006 0.022 0.035 0.097 0.110 0.155 0.229 0.124 0.215
##
## For the frequency table, variable is rounded to the nearest 0
## -----
## sci3
##          n missing distinct      Info      Mean      Gmd      .05      .10
##      9216         0         10      0.983      5.522      2.733         1         2
##      .25      .50      .75      .90      .95
##          4         6         7         9         10
##
## Value          1         2         3         4         5         6         7         8         9         10
## Frequency      584      382      997      1231      1322      1803      837      880      501      679
## Proportion 0.063 0.041 0.108 0.134 0.143 0.196 0.091 0.095 0.054 0.074
##
## For the frequency table, variable is rounded to the nearest 0
## -----
## sci4
##          n missing distinct      Info      Mean      Gmd      .05      .10
##      9216         0         10      0.983      5.896      2.701         2         3
##      .25      .50      .75      .90      .95
##          4         6         8         9         10
##
## Value          1         2         3         4         5         6         7         8         9         10
## Frequency      427      255      854      1122      1241      1877      957      1041      561      881
## Proportion 0.046 0.028 0.093 0.122 0.135 0.204 0.104 0.113 0.061 0.096
##
## For the frequency table, variable is rounded to the nearest 0
## -----

```

```
## sci5
##      n missing distinct      Info      Mean      Gmd      .05      .10
##    9216      0      10    0.975    7.254    2.653      3      4
##      .25      .50      .75      .90      .95
##      6      8      9      10      10
##
## Value      1      2      3      4      5      6      7      8      9      10
## Frequency  158  137  441  690  751  1143  1018  1518  1148  2212
## Proportion 0.017 0.015 0.048 0.075 0.081 0.124 0.110 0.165 0.125 0.240
##
## For the frequency table, variable is rounded to the nearest 0
## -----
## sci6
##      n missing distinct      Info      Mean      Gmd      .05      .10
##    9216      0      10    0.968    7.588    2.037      5      5
##      .25      .50      .75      .90      .95
##      6      8      9      10      10
##
## Value      1      2      3      4      5      6      7      8      9      10
## Frequency   59   42  130  223  830  1058  1724  2206  1152  1792
## Proportion 0.006 0.005 0.014 0.024 0.090 0.115 0.187 0.239 0.125 0.194
##
## For the frequency table, variable is rounded to the nearest 0
## -----
## eco_sum
##      n missing distinct      Info      Mean      Gmd      .05      .10
##    9216      0      36    0.994 2.494e-16    1.1 -1.65899 -1.24748
##      .25      .50      .75      .90      .95
## -0.63020 -0.01293  0.60435  1.42738  1.63314
##
## lowest : -4.1281 -3.92234 -3.71658 -3.30506 -3.0993
## highest:  2.45617  2.66193  2.86769  3.07345  3.27921
## -----
## sci_sum
##      n missing distinct      Info      Mean      Gmd      .05
##    9216      0      50    0.998 -1.997e-16    1.132 -1.37505
##      .10      .25      .50      .75      .90      .95
## -1.21654 -0.83612 -0.07528  0.68556  1.31959  1.82682
##
## lowest : -4.51351 -3.87948 -3.75267 -3.62586 -3.37225
## highest:  1.82682  1.95363  2.08043  2.20724  2.33405
## -----
## eco_sum_cw
##      n missing distinct      Info      Mean      Gmd      .05      .10
##    9216      0      191      1 -7.54e-18    1.103 -1.6311 -1.1430
##      .25      .50      .75      .90      .95
## -0.5850 -0.1230  0.6059  1.3234  1.7315
##
## lowest : -4.52775 -4.42219 -4.18962 -3.98496 -3.70865
## highest:  3.2317  3.44373  3.44793  3.59537  3.59704
## -----
## sci_sum_cw
##      n missing distinct      Info      Mean      Gmd      .05
##    9216      0      246      1 -1.215e-16    1.122 -1.51166
##      .10      .25      .50      .75      .90      .95
```

```

##      -1.14441   -0.71476   -0.06969    0.63854    1.34934    1.74080
##
## lowest : -4.86805 -4.62271 -4.4384  -4.29518 -4.16346
## highest: 3.11343  3.50677  3.90012  4.29346  4.49014
## -----
## edu_num
##      n missing distinct    Info    Mean    Gmd
##    9161      55        4    0.871    2.617    1.148
##
## Value      1      2      3      4
## Frequency  1167  4137   895  2962
## Proportion 0.127 0.452 0.098 0.323
##
## For the frequency table, variable is rounded to the nearest 0
## -----
## marital_weight
##      n missing distinct    Info    Mean    Gmd
##    9190      26        2    0.646    0.9373   0.0861
##
## Value      0.8    1.0
## Frequency  2881  6309
## Proportion 0.313 0.687
## -----
## index
##      n missing distinct    Info    Mean    Gmd    .05    .10
##    8941      275      136      1    46.35    18.7    19.2    22.4
##      .25    .50    .75    .90    .95
##    34.0    47.0    59.0    67.0    71.0
##
## lowest : 15.2 16    16.8 17.6 18.4, highest: 90    92    93    94    97
## -----
## life_exp
##      n missing distinct    Info    Mean    Gmd    .05    .10
##    9215      1      12    0.988    81.7    4.501    75.03    75.03
##      .25    .50    .75    .90    .95
##    79.70    80.80    85.70    87.32    87.61
##
## Value      75.034 77.550 79.700 80.573 80.800 81.250 82.260 84.050 85.700
## Frequency    1323   594   607   1596   920   426   431   627   638
## Proportion  0.144 0.064 0.066 0.173 0.100 0.046 0.047 0.068 0.069
##
## Value      86.700 87.320 87.610
## Frequency    1082   416   555
## Proportion  0.117 0.045 0.060
##
## For the frequency table, variable is rounded to the nearest 0
## -----
## age_sq
##      n missing distinct    Info    Mean    Gmd    .05    .10
##    8998      218      76      1    2395    1716    441    576
##      .25    .50    .75    .90    .95
##    1089    2116   3481   4489   5041
##
## lowest : 324 361 400 441 484, highest: 7921 8281 8464 8649 9025
## -----

```

```
## age_pct
##      n missing distinct      Info      Mean      Gmd      .05      .10
##    8998      218      774        1    56.78    22.67    25.53    29.66
##      .25      .50      .75      .90      .95
##    40.45    57.09    71.97    83.28    88.06
##
## lowest : 20.5456 20.6138 20.7612 21.0035 21.4158
## highest: 109.607 110.149 112.624 115.099 117.574
## -----
## eco_scaled_comp
##      n missing distinct      Info      Mean      Gmd
##    9216        0        3    0.816    0.6285    0.6824
##
## Value      0      1      2
## Frequency  4367  3906   943
## Proportion 0.474 0.424 0.102
##
## For the frequency table, variable is rounded to the nearest 0
## -----
## sci_Scaled_comp
##      n missing distinct      Info      Mean      Gmd
##    9216        0        3    0.743    0.5892    0.8048
##
## Value      0      1      2
## Frequency  5750  1502  1964
## Proportion 0.624 0.163 0.213
##
## For the frequency table, variable is rounded to the nearest 0
## -----
## outlier
##      n missing distinct
##    9216        0        2
##
## Value      mixed    pos
## Frequency  9215      1
## Proportion    1      0
## -----
```

```
#tidyverse
glimpse(subset_EA)
```

```
## Rows: 9,216
## Columns: 30
## $ country      <chr> "China", "China", "China", "China", "China", "China", ...
## $ sex          <chr> "female", "female", "male", "male", "male", "female", ...
## $ age          <dbl> 38, 69, 48, 70, 30, 27, 60, 63, 69, 68, 38, 26, 28, 65...
## $ marital_status <fct> partnered, single, partnered, partnered, partnered, si...
## $ education    <fct> NA, secondary, NA, tertiary, tertiary, tertiary, secon...
## $ eco1         <dbl> 4, 1, 7, 5, 8, 10, 1, 3, 2, 2, 7, 8, 7, 9, 6, 6, 6, 6,...
## $ eco2         <dbl> 6, 1, 7, 4, 8, 10, 9, 6, 2, 2, 9, 8, 8, 2, 8, 8, 4, 3,...
## $ eco3         <dbl> 1, 1, 8, 1, 8, 10, 1, 6, 3, 2, 2, 8, 9, 2, 5, 2, 5, 7,...
## $ eco4         <dbl> 10, 10, 8, 10, 3, 10, 8, 9, 8, 10, 9, 10, 9, 8, 9, 9, ...
## $ eco5         <dbl> 1, 1, 2, 1, 7, 1, 9, 2, 2, 1, 5, 1, 2, 3, 6, 2, 1, 7, ...
## $ eco6         <fct> environment, environment, environment, environment, en...
## $ sci1         <dbl> 9, 10, 10, 10, 8, 10, 5, 9, 6, 9, 9, 5, 9, 10, 10, 10,...
## $ sci2         <dbl> 9, 10, 10, 10, 4, 10, 5, 5, 10, 9, 9, 7, 9, 10, 10, 10...
## $ sci3         <dbl> 8, 1, 8, 1, 6, 4, 6, 5, 6, 2, 9, 6, 4, 1, 10, 9, 9, 4,...
## $ sci4         <dbl> 8, 1, 9, 1, 3, 4, 1, 5, 6, 9, 9, 6, 8, 1, 10, 7, 6, 3,...
## $ sci5         <dbl> 8, 1, 10, 10, 9, 10, 9, 8, 2, 2, 10, 6, 10, 10, 10, 6,...
## $ sci6         <dbl> 8, 10, 10, 10, 8, 10, 8, 9, 10, 10, 10, 8, 8, 10, 10, ...
## $ eco_sum      <dbl> -0.63020231, -2.27627012, 1.22162397, -0.83596079, 0.6...
## $ sci_sum      <dbl> 1.06598176, -1.08973048, 1.95362798, 0.05152894, -0.45...
## $ eco_sum_cw   <dbl> -0.50003369, -2.00807796, 1.19651611, -0.68853923, 0.6...
## $ sci_sum_cw   <dbl> 0.574191952, -1.860493789, 1.576709610, -0.571542514, ...
## $ edu_num      <dbl> NA, 2, NA, 4, 4, 4, 2, 2, 2, 2, 4, 4, 2, 2, 4, NA, 2, ...
## $ marital_weight <dbl> 1.0, 0.8, 1.0, 1.0, 1.0, 0.8, 1.0, 1.0, 1.0, 1.0, 1.0, ...
## $ index        <dbl> NA, 56.8, NA, 74.0, 34.0, 24.8, 62.0, 65.0, 71.0, 70.0...
## $ life_exp     <dbl> 80.573, 80.573, 75.034, 75.034, 75.034, 80.573, 80.573...
## $ age_sq       <dbl> 1444, 4761, 2304, 4900, 900, 729, 3600, 3969, 4761, 46...
## $ age_pct      <dbl> 47.16220, 85.63663, 63.97100, 93.29104, 39.98187, 33.5...
## $ eco_scaled_comp <dbl> 0, 0, 0, 0, 1, 0, 0, 2, 0, 0, 1, 0, 0, 0, 1, 1, 1, 0, ...
## $ sci_Scaled_comp <dbl> 0, 1, 0, 2, 2, 0, 1, 2, 2, 2, 0, 2, 0, 2, 0, 0, 0, 1, ...
## $ outlier      <fct> mixed, mixed, mixed, mixed, mixed, mixed, mixed, mixed...
```